23. Chloroform reacts with nitric acid to produce

a) nitro toluene

b) nitro glycerine

c) chloropicrin

d) chloropicric acid

24. acetone $\xrightarrow[\text{ii) } H_2O / H^{-1}]{\text{i) } CH_3MgI}$ X, X is

a) 2-propanol

b) 2-methyl-2-propanol

c) 1-propanol

d) acetonol

25. Silverpropionate when refluxed with Bromine in carbontetrachloride gives

a) propionic acid

b) chloro ethane

c) bromo ethane

d) chloro propane

26. Classify the following compounds in the form of alkyl, allylic, vinyl, benzylic halides

a) $CH_3 - CH = CH - Cl$

(ii) $C_6H_5CH_2I$

(iii) $\mathbf{CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3}$

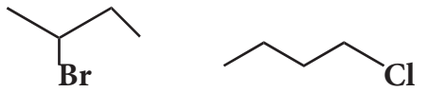
(iv) $CH_2 = CH - Cl$

II. Write brief answer to the following questions.

27. Why chlorination of methane is not possible in dark?

28. How will you prepare n propyl iodide from n-propyl bromide?

29. Which alkyl halide from the following pair is i) chiral ii) undergoes faster S_N2 reaction?



30. How does chlorobenzene react with sodium in the presence of ether? What is the name of the reaction?

31. Give reasons for polarity of C-X bond in halo alkane.

32. Why is it necessary to avoid even traces of moisture during the use of Grignard reagent?

33. What happens when acetyl chloride is treated with excess of CH_3MgI?

34. Arrange the following alkyl halide in increasing order of bond enthalpy of RX

CH_3Br, CH_3F, CH_3Cl, CH_3I

35. What happens when chloroform reacts with oxygen in the presence of sunlight?

36. Write down the possible isomers of $C_5H_{11}Br$ and give their IUPAC and common names.

37. Mention any three methods of preparation of haloalkanes from alcohols.

38. Compare S_N1 and S_N2 reaction mechanisms.

39. Reagents and the conditions used in the reactions are given below. Complete the table by writing down the product and the name of the reaction.

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